

NU Racing's 2025 MCHA Team Scopes/Projects – 1 FYP

This document contains the preferred amount of FYP Scopes and Directed Projects for NU Racing 2025's team.

Assumptions:

- Number of FYPs: 0
- Number of Directed Reading/Extracurricular: 4

FYP Scope contain:

- Starter project
- Scope for the year

Directed Reading/Extracurricular Projects contain:

- Scope for the project

1. FYP Scopes and Starter Projects

Accumulator

- New AIL
 - TASKS:
 - Implement changes from Ivo's report
 - Breakout board mounting style
 - LTSpice simulation to validate circuit design
 - DELIVERABLE:
 - AIL ordered (with LTSpice or breadboard test)
- Fix AMS Fault
 - TASKS:
 - Replace faulty CANaMons (CANaMon 1)
 - If CANaMon replacement doesn't work
 - Investigate BMS functionality
 - DELIVERABLE:
 - AMS fault no longer occurs
- Top plate fuse holders (maybe?)
 - TASKS:
 - Change to fuse holders that are RIGHT on the edge of a PCB, so that the wire is short enough to not need a fuse
 - DELIVERABLE:
 - New solution implemented
- New LVD
 - TASKS:
 - Interpose from LVD to CEN
 - Use of IRLB3813 MOSFET for cooling circuit (gather information from CEN engineer) or use automotive relay
 - Investigate high current draw of accum fans
 - DELIVERABLE:
 - New LVD ordered
- NU26 Accumulator
 - TASKS:
 - Design new voltage tap and temperature monitoring for pouch cell accumulator
 - Pouch cell testing and validation (maybe)
 - DELIVERABLE:
 - Voltage taps and temperature monitoring designed

AV.One – FYP – Jayden Horvath

- Starter Project
 - Read Lena's and Sebastian Scott's reports
 - 'Past_Reports' folder within 'Racing-AV1' GitHub.
 - LVD rework to enable use of charger with AV.One
 - Troubleshoot DCDC
 - Deliverable:
 - New LVD ordered and found issue with DCDC
- Implement and test 2x Raspberry HQ Cameras for stereo vision
- Investigate use of either Marine's LiDAR or use of spare LiDAR that is somewhere on campus – refer to Malcolm for details
- Attempt to implement as many Autonomous Advancements as possible from Lena's Report (section 8.2.)
 - Contact UQ's autonomous race team on issues encountered and/or direction
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2. Directed Reading Scopes

- PEN/BSPD and potentially TSAL_Discharge – Directed – Jayson
 - Where to start:
 - Read Bushy's reports the BSPD and the TSAL_Discharge breakout boards
 - Lukes' report on why the resistor voltage divider was a time-consuming idea. With too many variables to account for during the commissioning process.
 - Read Alec Chapman's ENGG4200 Report on the topology of the nodes
 - Read the section of PEN from Joshua Hayward's report, to gain a brief understanding of the PEN
 - Read Josh Dawson's report
 - Mainly:
 - 4.4 Proposed Modifications
 - 4.5 CAD/Schematic/PCB/Code
 - 4.6.3 Charger V1 PCB
 - 4.7.1 ECU Verification
 - BSPD
 - Remove voltage divider for C_REF. Replace with potentiometer, as this is easier for someone to understand how to tune it correctly
 - TSAL_Discharge
 - Change current naming of red to amber on Bushy's V3 and order
 - Potentially implement green light turning off while red is flashing
 - DEN:
 - Resize to chassis (might not have to?)
 - All naming of AMS to BMS
 - Change from DT 8 pin to AT 8 pin keyed connector for CEN connection
 - PEN (later in semester, waiting for new Pedal Box to be designed)
 - Fit to pedal box
 - Add capacitors to brake sensor inputs
 - Investigate need for Zener diodes, and if possibly removing them
 - Charger
 - Add new HV connectors
 - Fix oopsie wire on PCB
 - In charge of monitoring and managing MoTeC at track days
- HIP – Directed or Extra – Mia (is ENGG3200 in SEM 2)
 - Read Jackson's report?
 - All naming of HVD to MSD
 - New Master Switches (footprint will change)
 - HV connector interlock (part of the mech enclosure role?)

- Change from DT connector to one that is rated for higher voltage for DCDC to HIP connector

Tasks To Be Completed During The Year

- Motor Controller
 - TASKS:
 - Hayward's recommendations
 - Implement pack current derating at cell/pack temperatures over 50 degrees.
 - PEN code
 - Use either maxsegtemp or mincellvol/maxcellvolt
 - Derate from the 3 degrees under the max temperature rating of the VTC6
 - Limit current down to ~30A under high temperatures. Shut down all current above 60 degrees
 - Other recommendations from his report
 - DELIVERABLE:
 - Pack current derating implemented
 - Other recommendations from Hayward's report also implemented
- Dynamometer
 - TASKS:
 - Create Load Testing Node
 - Have it be very similar to the CEN from NU23
 - Read Dawson's report on the dyno
 - Implement as much as possible from this report
 - DELIVERABLE:
 - If all goes well, (which it won't), we will be able to:
 - Map power curves
 - Tune the motor
- Launch Control
 - TASKS:
 - Use wheel speed sensors (implemented through another task) to implement launch control on the PEN
 - DELIVERABLE:
 - Launch control implemented

- HFR PCB Iteration
 - TASKS:
 - Implement BMS-IMD Tractive System Status Indicator fault detection circuit on this board
 - Read 2025 rules and compare them to 2024's rules to understand the change
 - Research circuit to implement this change
 - Design circuit
 - AND gate for BMS and IMD fault detection
 - 555 timer circuit for 2-5 Hz signal generation (research correct circuit (555 timer datasheet), but ultimately use the one that is on the TSAL_Discharge board (use DMN6140L for all MOSFETs)
 - MOSFET to switch outputted GND signal
 - Rough outline of circuit in obsidian notes
 - LTSpice/Breadboard the circuit
 - Breadboard the circuit
 - Red Flash of TSSI could be just a THT LED?
 - Correct PCB to accommodate for circuit
 - Increase input/output of header pins by 1 on each side
 - Implement green light turning off while red is flashing
 - Potentially make a second circuit to account for this possibility
 - Rough outline of circuit in obsidian notes
 - HFR PCB Ordered
 - DELIVERABLE:
 - HFR PCB manufactured and commissioned with changes detailed above

- CEN Iteration
 - TASKS:
 - Fit to opposite side of chassis
 - TSAL DT
 - Change name to UEN
 - Make it an 8-pin connector (to route new red switch ground signal)
 - Keyed AT 8 pin connectors for DCDC, DEN, REAR, MOTOR CONTROLLER
 - Route TSAL_GREEN_SWITCHED_GND (rename) to HFR breakout
 - Interpose on CEN from LVD
 - IRLB3813 MOSFET on cooling circuits (or have another double switching MOSFET)
 - MOTHER Board iteration ordered
 - DELIVERABLE:
 - New MOTHER Board iteration manufactured and commissioned with changes detailed above

- Ready To Move Light Implementation
 - TASKS:
 - Order and commission a light that meets DOT FMVSS 108
 - Change loom to route an extra wire up to the Ready To Move Light
 - Design mounting for 2x amber DOT FMVSS 108 lights on the car (one facing forward and one facing backwards (aft))
 - DELIVERABLE:
 - Ready To Move Light mounted
 - Loom has changed to account for extra signal

- EXPANSION Board PCB Iteration
 - TASKS:
 - Size CAN lines correctly
 - Fix ground plane (not have big breaks)
 - Transfer SWEN functionality
 - Includes: dials for torque setting, switch for regen (implement a way so that the dial/switch are in the correct position after unplugging and plugging the board in)
 - Temperature sensors for radiator
 - Create loom to connect to them (already in radiator)
 - DBC file and MoTeC updates
 - Sensors for the board to work with:
 - brake temperature
 - shock pots
 - wheel speed

- steering angle
 - EXPANSION Board Ordered
- DELIVERABLE:
 - EXPANSION Board manufactured and commissioned with changes detailed above
- EXPANSION Board Integration
 - TASKS:
 - Brake temperature sensors able to work over MoTeC
 - Loom to connect sensors to EXPANSION Board
 - Create 4 messages in NU25 DBC file for wheel speed sensors (one for each wheel), with correct scaling, offset, bit length and value range
 - Implement code for brake temperature sensors on the EXPANSION Board
 - Configure MoTeC to read and display new message
 - DELIVERABLE:
 - Brake temperature sensors able to display data over MoTeC